

Operational Plan v2: Locomotive Digital Twin Real-Time Telemetry & Health Index Prototype

Engineering Development Team

April 2026

1 Executive Summary

Build a **fullstack prototype** of a visual digital twin dashboard for KTZ locomotives. The system ingests real-time telemetry via WebSocket/SSE, computes a composite **Health Index (0–100 / A–E)**, and renders it in a 3D-enhanced interactive dashboard with route context, alerts, and historical replay.

Core deliverables (mapped to judging weights):

- 1. Real-time Engine (35%) — WebSocket/SSE pipeline, highload resistance (x10 spikes), fault tolerance, j500 ms latency.
- 2. Frontend & UX (30%) — signal prioritization, informative charts, 3D twin, accessibility.
- 3. Backend & Architecture (25%) — clean modularity, history storage, scalable Health Index logic.
- 4. Engineering Culture (10%) — Git hygiene, launch instructions, OpenAPI docs, logs/metrics, healthcheck.

2 Target Locomotive Scope

The prototype targets two primary locomotive classes to demonstrate both electric and diesel-electric digital twins:

Feature	KZ8A (Electric)	TE33A (Diesel-Electric)
Power Source	25 kV Catenary	GEVO-12 Diesel Engine
Power Output	8.8 MW	3.3 MW
Control System	Microprocessor (IGBT)	CCA Architecture
Anti-Slip	Predictive/Real-time	Reactive Response
Operating Range	−50°C to +50°C	−50°C to +50°C
Telemetry Params	19 core parameters	25 core parameters

Table 1: Primary locomotive types for the prototype.

Legacy units (VL80, 2TE116) and variants (TEP33A, KZ4AT, 2TE25KM) are modeled as future extensions via a pluggable data schema.

3 Complete Telemetry Metrics Catalog

The dashboard monitors the following domains. Each metric feeds into the Health Index via its respective modifier.

Metric	Unit	Range	HI Modifier
Speed	km/h	0–200	M_{Ld}
Acceleration	m/s ²	−3 to +3	M_{Ld}
Tractive Effort	kN	0–500	M_{Ld}
Wheel Slip Ratio (ε)	%	0–15	M_H
Adhesion Coefficient (μ)	—	0.05–0.40	M_H
Traction Motor Current	A	0–1500	M_{Ld}
Traction Motor Torque	N·m	0–5000	M_{Ld}

Metric	Unit	Range	HI Modifier
Fuel Level (diesel)	%	0–100	M_R
Fuel Consumption Rate	L/h	0–600	M_{Ld}
Catenary Voltage (electric)	kV	19–29	M_H
Traction Circuit Voltage	V	0–3600	M_H
Electric Power Consumption	kW	0–8800	M_{Ld}
Battery Voltage	V	90–140	M_R
Auxiliary Power Load	kW	0–200	M_{Ld}

Metric	Unit	Range	HI Modifier
Engine Oil Temperature	°C	40–130	M_H
Coolant Temperature	°C	40–110	M_H
Engine Oil Pressure	MPa	0.2–0.8	M_H
Exhaust Gas Temperature	°C	200–700	M_H
Turbocharger RPM	RPM	0–80000	M_H
Compressor Discharge Pressure	MPa	0.5–1.0	M_R
Traction Motor Winding Temp	°C	20–180	M_H
Transformer Oil Temp (electric)	°C	20–100	M_H
Vibration Amplitude (wheelset)	mm/s	0–50	M_H
Ambient Temperature	°C	−50 to +50	M_H

Metric	Unit	Range	HI Modifier
Main Reservoir Pressure	MPa	0.7–0.9	M_R
Brake Cylinder Pressure	MPa	0–0.4	M_R
Brake Pad Wear	% remaining	0–100	M_R
Solenoid Valve Residual Signal	mV	0–500	M_R
Parking Brake Status	bool	0/1	—

3.1 Traction & Motion

3.2 Power & Resources

3.3 Subsystem Health (Thermal & Mechanical)

3.4 Braking Systems

3.5 Reliability & Operational KPIs

3.6 Infrastructure & Environment

4 Health Index Engine

4.1 HI Formula

$$HI(t) = \left[HI_{\text{init}} \cdot e^{M_{Ld}(t) \cdot \beta \cdot t} \right] e^{(M_H(t) + M_R(t))} \quad (1)$$

- HI_{init} : Baseline health after last major service (set to 100).

Metric	Unit	Source	HI Modifier
Active Error Codes	count	ECU/PLC	M_R
Error Code Frequency	events/h	aggregated	M_R
Operating Hours Since Last Service	h	maintenance log	M_R
Mean Time Between Failures (MTBF)	h	historical	M_R
Mean Time To Repair (MTTR)	h	historical	M_R
Locomotive Availability	%	calculated	M_R
Distance Since Last Overhaul	km	odometer	M_R

Metric	Unit	Source	HI Modifier
GPS Position (lat, lon)	degrees	GPS module	—
Track Gradient	‰	route DB	M_{Ld}
Speed Limit (current segment)	km/h	route DB	—
Vertical Dynamics Coefficient	—	accelerometer	M_H
Frame Force	kN	strain gauge	M_H
Rail Surface State	enum	inferred	M_H

- β : Base aging rate (calibrated per locomotive type).
- $M_{Ld}(t)$: **Load Modifier** — aggregated operational stress (tractive effort, power, fuel burn, acceleration).
- $M_H(t)$: **Health Modifier** — thermal, vibration, friction, and environmental signals.
- $M_R(t)$: **Reliability Modifier** — maintenance history, error codes, brake wear, MTBF.

4.2 Output Format

- Numeric score: 0–100.
- Letter grade: A (≥ 85), B (≥ 70), C (≥ 50), D (≥ 30), E (< 30).
- **Factor breakdown**: dashboard shows top contributing factors (e.g., “Oil temp +12°C above norm → −8 points”).
- Trend sparkline: HI over last 24h / 7d / 30d.

4.3 Alert Thresholds

Level	HI Range	Action
Normal	85–100 (A)	No action
Advisory	70–84 (B)	Schedule inspection
Caution	50–69 (C)	Prioritize maintenance
Warning	30–49 (D)	Limit operations
Critical	0–29 (E)	Immediate intervention

5 Technical Architecture

5.1 Stack

5.2 Data Flow

```

Telemetry Simulator --> FastAPI Ingest Endpoint (POST /telemetry)
                        |
                        +--> Redis Pub/Sub (buffer spikes)
                        |
                        +--> PostgreSQL (persist history)
                        |
                        +--> HI Engine (compute score)

```

Layer	Technology
Backend API	FastAPI (Python 3.11+), async
Real-time Transport	WebSocket + SSE fallback
Database	PostgreSQL (time-series via TimescaleDB or partitioned tables)
Cache / Pub-Sub	Redis (event buffering, spike absorption)
3D Visualization	React Three Fiber (Three.js)
Charts & Graphs	Recharts / Visx
Route Map	Leaflet / Mapbox GL JS
Auth	JWT-based, basic login
API Docs	Auto-generated OpenAPI / Swagger UI
Observability	Prometheus + Grafana + Loki
Containerization	Docker Compose (single-command launch)

```

|
+--> WebSocket/SSE --> React Dashboard
|
+--> Alert Engine --> Notification channel

```

5.3 Highload Resistance Strategy

- Redis as event buffer absorbs x10 burst spikes before DB write.
- Batch inserts to PostgreSQL (configurable window: 100ms–1s).
- WebSocket backpressure: drop oldest frame if client lags.
- Telemetry simulator includes a “spike mode” toggle for demo.
- Frontend uses `requestAnimationFrame` throttling to prevent UI freeze.

5.4 Fault Tolerance

- SSE fallback if WebSocket connection drops.
- Reconnect logic with exponential backoff on the client.
- Healthcheck endpoint: `GET /health` returns service status + DB connectivity.
- Graceful degradation: dashboard shows last-known-good state with “stale” indicator.

6 Security & Engineering Standards

- All secrets (DB password, JWT secret) via `.env` / environment variables.
- Basic JWT authentication; role-based access (operator vs. admin).
- CORS whitelist for frontend origin.
- OpenAPI/Swagger auto-generated at `/docs`.
- Structured JSON logging (Loki-compatible).
- `/health` and `/metrics` (Prometheus) endpoints.
- Clean Git history: conventional commits, `README.md` with launch instructions.
- `docker-compose up` single-command bootstrap.

7 Dashboard Modules

1. **Health Index Gauge** — large central dial (0–100, color-coded A–E), with factor breakdown panel.
2. **3D Locomotive Twin** — React Three Fiber model with color-coded subsystem overlays (green/yellow/red per component).
3. **Traction & Motion Panel** — real-time speed, tractive effort, and acceleration charts with trend lines.
4. **Resource Panel** — fuel level gauge, consumption rate, voltage/power indicators.
5. **Subsystem Monitoring** — temperature and pressure gauges with threshold markers and alerts.
6. **Braking Panel** — reservoir pressure, cylinder pressure, pad wear indicators.
7. **Alert Feed** — live stream of warnings/errors with severity, timestamp, and error code.

8. **Route Map** — interactive map (Leaflet) showing current position, speed limits, and track gradient.
9. **History & Replay** — select time range, replay telemetry session, export data as CSV/JSON.
10. **Fleet Overview** — (stretch goal) grid view of multiple locomotives with HI summary.

8 Evaluation Metrics

8.1 Model & Engine Accuracy

- **MARE** — Mean Absolute Relative Error for HI prediction drift.
- **RMSE** — Root Mean Square Error for HI variance analysis.
- **AUROC** — Area Under ROC for fault classification (normal vs. degraded).
- **Precision / Recall** — for alert triggering (false positive rate vs. missed faults).
- **F1-Score** — harmonic mean of precision and recall for balanced evaluation.
- **Monotonicity** — HI should decrease monotonically under sustained stress (no false recoveries).
- **Trendability** — correlation between HI trajectory and actual degradation curve.

8.2 System Performance

- **End-to-end Latency** — event generation to dashboard render < 500 ms (P95).
- **Throughput** — sustain ≥ 1000 events/sec per locomotive under x10 spike.
- **UI Frame Rate** — ≥ 30 FPS during live streaming (no jank).
- **Reconnect Time** — WebSocket recovery < 3 s.
- **DB Write Latency** — batch insert P99 < 200 ms.

8.3 Operational KPIs (displayed on dashboard)

- Locomotive Availability (%).
- MTBF / MTTR.
- Fuel efficiency (km/L or kWh/km).
- Wheel wear rate (mm/10⁴ km).
- Brake pad remaining life (%).
- Maintenance cost per operating hour (stretch goal).

9 Implementation Phases

9.1 Phase 1: Foundation (Day 1, Hours 0–6)

1. Initialize monorepo: `/backend`, `/frontend`, `/simulator`, `docker-compose.yml`.
2. FastAPI skeleton with `/health`, `/docs`, and CORS.
3. PostgreSQL schema: `locomotives`, `telemetry_events`, `alerts` tables.
4. Telemetry simulator (Python script) generating realistic multi-param events.
5. `POST /telemetry` ingest endpoint with Redis pub/sub fan-out.
6. WebSocket endpoint streaming latest events to connected clients.
7. React app scaffold with WebSocket connection hook.

9.2 Phase 2: Health Index & Core UI (Day 1, Hours 6–12)

1. Implement HI calculation engine (formula + modifier weights).
2. HI API: `GET /locomotives/{id}/health` with factor breakdown.
3. Dashboard layout: HI gauge, traction panel, resource panel.
4. Real-time chart components (speed, temperatures, pressures).
5. Alert engine: threshold checks → persist to DB → push via WebSocket.
6. Alert feed component with severity badges and timestamps.

9.3 Phase 3: 3D Twin & Advanced Features (Day 2, Hours 12–20)

1. 3D locomotive model in React Three Fiber with subsystem color overlays.
2. Route map (Leaflet) with GPS position, speed limits, gradient.

3. History API: GET /telemetry?from=&to=&locomotive_id= with pagination.
4. Replay mode: frontend plays back historical session at adjustable speed.
5. Export endpoint: GET /export?format=csv&from=&to=.
6. JWT authentication: login page, protected routes, ENV secrets.

9.4 Phase 4: Hardening & Demo Prep (Day 2, Hours 20–24)

1. Highload stress test: run simulator in spike mode, verify no crashes.
2. Latency measurement: instrument P95 end-to-end.
3. Prometheus metrics endpoint + Grafana dashboard (optional).
4. README with architecture diagram, setup instructions, and screenshots.
5. Record demo scenario: normal → degradation → alert → recovery.
6. Final Git cleanup and tag release.

10 Wheel-Rail Friction Integration (Stretch Goal)

Friction and adhesion analysis feeds into M_H and M_{Ld} :

- Slip ratio: $\varepsilon = (V_L - V_W)/V_L$; alert if $\varepsilon > 3\%$.
- Rail surface classification: clean / wet / oily / sanded (inferred from μ curves).
- FFT vibration analysis for friction signatures and lubrication needs.
- Expected impact: 10–15% reduction in wheel/rail wear; 10% longer maintenance intervals.

11 Key References

1. Chen, Z. et al. (2022). Deep learning feature fusion based health index construction. *IEEE Trans. Reliability*.
2. Alstom Transport. (2012). *KZ8A: To operate in extreme conditions*.
3. GE Transportation. (2018). *Operating manual, TE33AC* (GEJ-7227).
4. Sansyzbay, K. et al. (2025). Anti-slip system modeling in wheel-rail contact. *IJMERR*, 14(6).
5. Abdullayev, S. (2024). TE33A brake equipment tests. *Comm. Univ. Zilina*, 26(4).
6. Gómez Fernández, J. F. et al. (2025). Asset health index integration in digital twins. *Energies*, 18(12).
7. ISO 13381-1:2015. Condition monitoring — Prognostics — General guidelines.
8. Davidyan, G. et al. (2023). Digital twin of locomotive braking system solenoid valve. *PHM Society*.
9. FlaBingo/flux-station — IIoT platform for 3D digital twins (FastAPI + R3F).
10. NatLabRockies/altrios — Advanced Locomotive Technology & Rail Infrastructure Optimization.